

Drone position and line of sight calculator

Overview —

Calculator needs to be able to work out the distance a drone is at and 'shoot' it down.

It will need the drones coordinates (x, y) and the ground station coordinates $(0, 0)$ and determine:

- Distance to drone
- Angle to 'shoot' it down

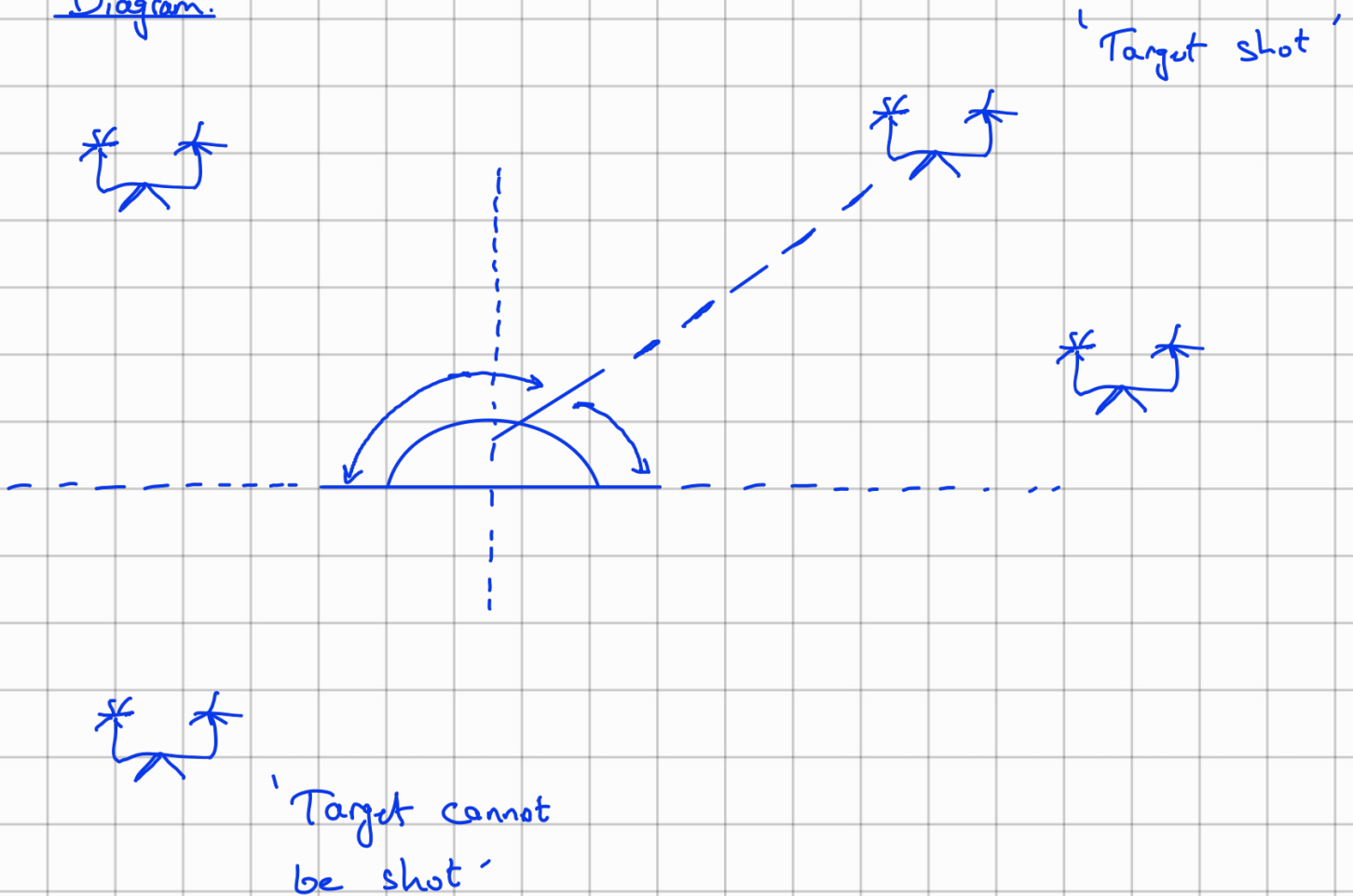
Then return

'Target down' or 'Target Missed/Unable to shoot'

Constraints:

- Ground station gun can only move/rotate 180°
- Drones are stationary. They 'spawn' in @ a given coordinate position (x, y) .

Diagram:



Assumptions:

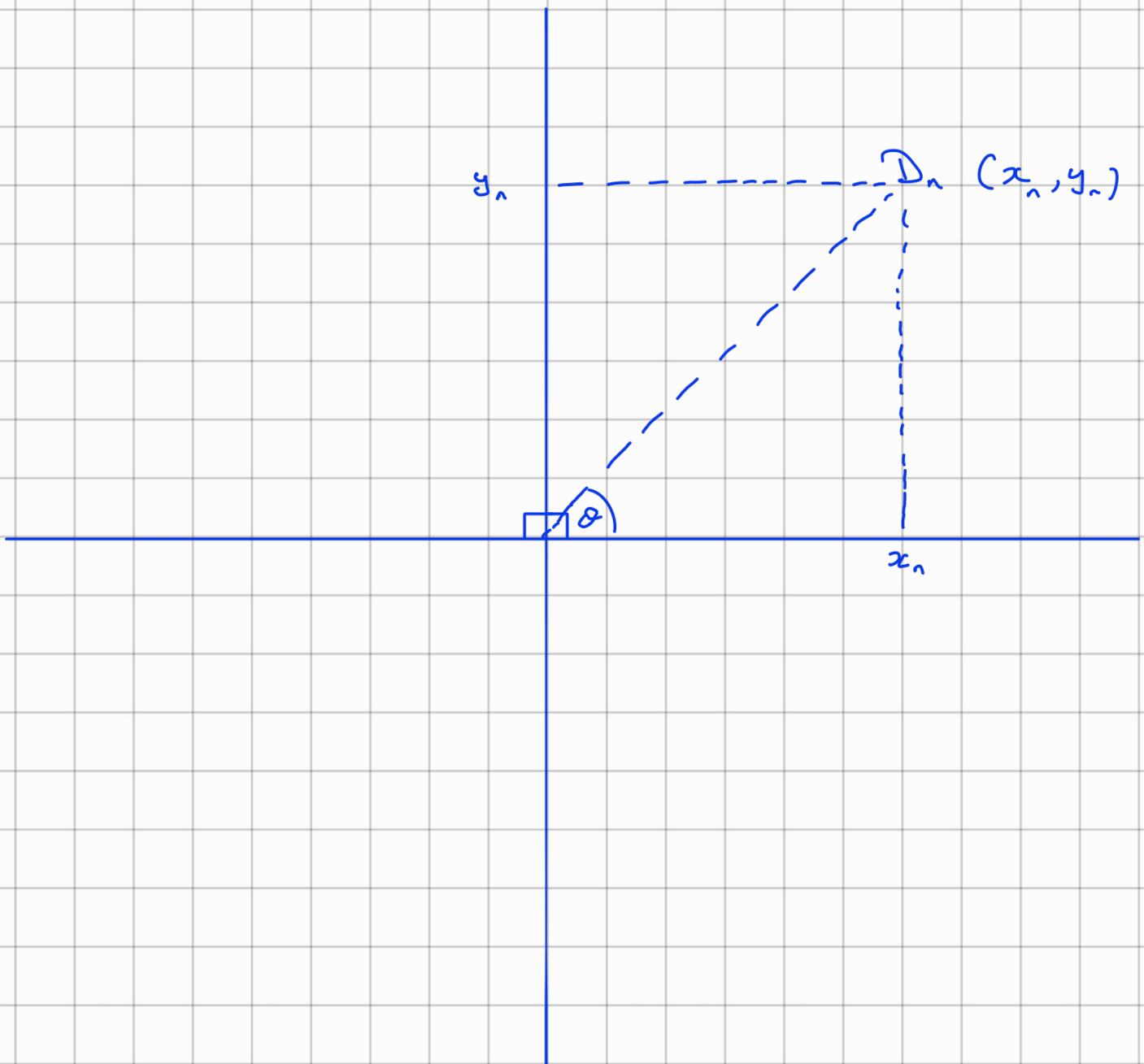
- 2D

- Ground station is flat and on 0,0
- Drones 'spawn' and stay stationary
- Ground station can only angle itself w/in 180°
- Drones can spawn anywhere

Equations:

Ground station is @ position $(0,0)$ ($\square$)

Drones are @ position (x_n, y_n) , D_r



From $(0,0) \rightarrow (x_n, y_n) :$

Distance : R

$$R = \sqrt{x^2 + y^2}$$

Theta : θ

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

